gemma4:31b on WorkBench templates, run in OfficeBench

English vs Bangla · 11 templates × 3 variants × 2 languages = 66 runs · run 2026-10-10 · tag `wb1` · generated by `wb_bench/report/build_report.py` from the 66 `result.json` files

Summary

30/33	31/33	33/33 · 32/33	0
English, strict (90.9%)	Bangla, strict (93.9%)	EN · BN, counting grader-assumption failures as passes	runs with a crash or harness error

- **All 66 runs completed without errors.** Strict score: English 30/33, Bangla 31/33.
- **Only one of the five failures is a real model mistake** (T15.v1, Bangla: it deleted two meetings with Dana instead of the next one). The other four come from my grader's assumptions: a 9:00 workday start (T16) and an exact event title (T20).
- **No English–Bangla difference is visible** on these 11 templates: the model is near the ceiling in both languages. This agrees with the earlier WorkBench study (gemma4:31b 81.1% English vs 82.2% Bangla on 90 tasks, McNemar p = 1.000).
- **One failure pattern repeats across both environments:** in WorkBench and here, gemma's English T16 runs ("book a catch-up at my first free slot") pick a different slot than expected, while the Bangla runs pick the expected one.
- **OfficeBench needed two fixes to run these tasks at all:** its `delete_event` action can never delete an event (two bugs), and my grader had to ignore the sent copy that `send_email` files in the sender's folder. Neither bug affects your earlier 300-task OfficeBench scores, because none of those tasks deletes a calendar event.

1. What was run

The Bangla translations in `batch_1..5.csv` are WorkBench templates (T01–T69). OfficeBench has only email (list, read, send to one recipient), calendar (create, delete by exact title, list; no attendees, no edit), Excel, Word, PDF, OCR, shell and an LLM app. It has no CRM, analytics or project-management app, so only 11 templates map onto OfficeBench's apps without workarounds. They were run as OfficeBench tasks:

Item	Setting
Templates	T03, T10, T13, T15, T16, T17, T18, T19, T20, T54, T55 (email and calendar; 3 phrasings each)
Languages	English template text and the Bangla translation from the CSVs; OfficeBench's Bangla system prompt (`lang=bn`) for Bangla runs
Model	gemma4:31b on ollama.com, temperature 0, at most 20 steps, one run per cell
Slot values	Chosen by me and identical in both languages (names Carlos, Dana, Lee, Priya; today Wed 2024-05-15 09:00; user Alice)
Starting state	A hand-built calendar of 15 events and 2 inbox emails, designed so each task has one correct end state
Scoring	Strict end-state check (`wb_bench/evaluate.py`): Alice's calendar must equal the expected set of (title, start, end), and the delivered emails must equal the expected ones (recipient, subject, body phrase). Any extra change fails the run.

> **Not the same benchmark as WorkBench.** WorkBench has its own sandbox and ground truth. Here the slot values, starting data and grader are mine, so the scores below are *not* directly comparable with WorkBench scores. Section 4 compares the patterns.

2. Results

Template	Task	English	Bangla
T03	Send an email (body + subject)	3/3	3/3
T10	Cancel my first meeting on a date	3/3	3/3
T13	Cancel the next {event} meeting	3/3	3/3
T15	Cancel my next meeting with {name}	3/3	2/3
T16	If not met {name} in N days, book 30-min catch-up	1/3	3/3
T17	Cancel meetings on a day after a time	3/3	3/3
T18	Cancel all future meetings with {name}	3/3	3/3
T19	Cancel all future {event} meetings	3/3	3/3
T20	Create a 1.5-hour event at a date/time	2/3	2/3
T54	If not met {name} in N days, send email	3/3	3/3
T55	If no meeting with {name} in next N days, send email	3/3	3/3
Total (strict)		**30/33**	**31/33**

Mean steps per run: English 5.0, Bangla 4.7.

3. The five failures

Run	What the model did	Expected	Verdict
T15.v1 BN	Deleted "Meeting with Dana" (May 16) **and** "Project review with Dana" (May 20)	Delete only the next one (May 16)	Real error: over-deletion
T16.v2 EN	Booked "Catch up with Carlos" on May 16, 08:00–08:30	First free slot from 09:00: 09:30–10:00	Grader assumption (workday start)
T16.v3 EN	Booked "catch-up" on May 16, 08:00–08:30	09:30–10:00	Grader assumption (workday start)
T20.v1 EN	Created "Roadmap review with Dana", May 22 14:00–15:30	Title exactly "Roadmap review"	Grader assumption (exact title)
T20.v1 BN	Same as English	Title exactly "Roadmap review"	Grader assumption (exact title)

- **T16:** the template says "my first free slot from tomorrow" but never says when the workday starts. 08:00 is a defensible reading; my grader assumed 09:00. All three Bangla T16 runs chose 09:30 and passed.
- **T20:** OfficeBench events have no attendee field, so adding the person's name to the title is a reasonable way to record "with Dana". Time and duration were correct in both languages.
- **T15.v1 BN** is the only failure where the action itself is wrong. The model treated "the next meeting with Dana" as every upcoming meeting with Dana. The other two Bangla phrasings of T15 passed.

> Counting the four grader-assumption failures as passes gives **English 33/33 and Bangla 32/33**. Either way, the difference between languages is one run or fewer.

4. Comparison with the earlier WorkBench study

Source: `WORKBENCH_FAILURE_ANALYSIS.pdf` (generated 2026-10-10; WorkBench pilot, 90 tasks, 15 per domain; English condition C0 vs Bangla condition C6; WorkBench's own end-state check; one run per cell at temperature 0). The WorkBench repository is not on this machine, so tasks can only be matched to templates through the task descriptions quoted in that report.

4.1 Overall and by domain (gemma4:31b, WorkBench)

Domain	English	Bangla	Relevance here
All 90 tasks	73/90 (81.1%)	74/90 (82.2%)	Gap −1.1 pp, 95% CI [−10.0, 7.8], McNemar p = 1.000
Email	15/15	14/15	T03 here: 3/3 and 3/3
Calendar	13/15	14/15	Most of the templates run here are calendar templates
Multi-domain	8/15	9/15	T54, T55 here: 3/3 and 3/3
gpt-oss:20b, all 90	53/90 (58.9%)	48/90 (53.3%)	Not run here

4.2 WorkBench tasks from the same templates

WorkBench task	Template	What the report found	What happened here
calendar:63 (gemma)	T16 (catch-up)	English failed: booked "catch-up" at 11:00, overlapping a 10:00–12:00 event; ground truth and Bangla chose 13:00. Coded general model weakness (same slot error in multi_domain:84 and :200).	Same pattern: English T16.v2/v3 booked 08:00; all Bangla T16 runs chose the expected 09:30.
calendar:38 (gpt-oss)	T13 (cancel the next X)	English deleted the next "Quarterly Sales Review" **and** a past one.	gemma deleted only the next "Team sync" in all 6 runs (6/6 pass).
calendar:84 (gpt-oss)	likely T18 (cancel all future meetings)	Deleted past meetings against "future" (English; Bangla "সামনে" likewise).	gemma kept the past events in all 12 T18/T19 runs (12/12 pass).

The one real failure here (T15.v1 Bangla, deleting more meetings than asked) is the same kind of error as gpt-oss's calendar:38 in WorkBench: a "next" or "future" constraint is not kept. In WorkBench this was coded general model weakness, because it also happens in English.

> **Agreement:** both environments show no measurable English–Bangla gap for gemma4:31b on email and calendar tasks. The genuine Bangla failures found in WorkBench (relative-date phrases such as "গত সপ্তাহে") come from templates that cannot run in OfficeBench (they need email dates or the analytics app), so this run could not test them.

5. What happened in the OfficeBench environment

5.1 Context: the earlier 300-task OfficeBench runs

Run	Passed
gemma4:31b, English (tag `think`)	132/300 (44.0%)
gemma4:31b, Bangla (tag `bn`)	103/300 (34.3%)

OfficeBench's own 300 tasks are mostly file tasks (Excel, Word, PDF, OCR) plus email and calendar, and their gap (about 10 points) is larger than on the WorkBench templates. Those runs are context only; they are not part of this report's 66 runs.

5.2 Problems found in OfficeBench, and fixes

Problem	Effect	Fix (this run only; `apps/` untouched)
`delete_event` wraps the title in `"..."`, which breaks inside the `bash -c` wrapper	"Vendor call" arrives as "Vendor"; nothing matches, yet the app reports "Successfully delete"	Quote it as `create_event` does
`delete_event` calls `calendar.subcomponent.remove`; the attribute is `subcomponents`	Every real delete raises an error and reports "Failed"; an existing event can never be deleted	Patched copy of the file copied into the container
`send_email` also saves a sent copy in the sender's folder	My first grader counted it as an extra email	Grader ignores the sender's own folder; all runs regraded
`list_events` takes `username`, while `create_event` / `delete_event` take `user`	A model that reuses the argument name gets "Malformed action" (seen once in a pilot)	Not changed (part of the benchmark)

> **Correction to an earlier claim.** I first said the `delete_event` bugs affected your earlier 300-task OfficeBench scores. They did not: a check of all 300 task texts found no task that deletes or cancels a calendar event (the "delete" tasks are about Excel rows and files). The bugs matter only for tasks like the WorkBench templates, where 8 of the 11 templates delete events.

5.3 Run history

- A pilot (T10.v1) failed in both languages and exposed the two `delete_event` bugs. Every T10 run made before the fix was deleted and rerun.
- One restart was wasted by my own mistake: the patched apps were copied to `/apps_patched` instead of `/apps`. Those runs were deleted and rerun.
- The machine rebooted at 04:19 because memory and disk ran out (C: dropped to 0.02 GB free; Windows paging file on C:). After the reboot, Docker Desktop would not start until stale socket files in `AppData\Local\Docker\run` and `docker-secrets-engine` were moved aside. The WSL swap file was moved to `D:\WSL` and the WSL VM capped at 4 GB.
- The run resumed at 05:26 and finished at 05:40. None of the 66 graded runs has a harness error; finished runs were never rerun.

6. Limits

- **Small and easy:** 33 runs per language on the 11 simplest templates; the model is near the ceiling, so the run cannot show a gap of a few points (WorkBench's 90-task interval already allows up to about 8 points).
- **One run per cell** at temperature 0; run-to-run variation is not measured.
- **My slot values, starting data and grader.** Different choices (for example a 08:00 workday) would change T16 and T20.

- **Translations:** the Bangla text is taken from the CSVs unchanged. It uses ASCII digits (for example "30 মিনিটের"), and person and event names stay in Latin script. Native-speaker review is still pending, as in the WorkBench study.

Appendix: all 66 runs

FAIL* = failed only because of a grader assumption (section 3). Task text is exactly what the model received.

Variant	Lang	Task text	Steps	Result
T03.v1	EN	Send an email to Carlos saying 'Can we move our call to Friday?' and title it 'Call reschedule'	3	PASS
T03.v1	BN	Carlos-কে একটা ইমেইল পাঠাও, তাতে লেখা থাকবে 'Can we move our call to Friday?', আর সাবজেক্ট দিও 'Call reschedule'	4	PASS
T03.v2	EN	I need to send an email to Carlos saying 'Can we move our call to Friday?' and title it 'Call reschedule'. Can you do that?	3	PASS
T03.v2	BN	Carlos-কে একটা ইমেইল পাঠানো দরকার, যেখানে লেখা থাকবে 'Can we move our call to Friday?' আর সাবজেক্ট হবে 'Call reschedule'। তুমি কি এটা করে দিতে পারবে?	4	PASS
T03.v3	EN	please send an email to Carlos saying 'Can we move our call to Friday?' and title it 'Call reschedule'	3	PASS
T03.v3	BN	দয়া করে Carlos-কে একটা ইমেইল পাঠিয়ে দাও, তাতে লিখবে 'Can we move our call to Friday?' আর সাবজেক্ট রাখবে 'Call reschedule'	4	PASS
T10.v1	EN	Cancel my first meeting on May 16	4	PASS
T10.v1	BN	16 মে তারিখে আমার প্রথম মিটিংটা বাতিল করো	4	PASS
T10.v2	EN	Delete my first meeting on May 16	4	PASS
T10.v2	BN	16 মে তারিখের আমার প্রথম মিটিংটা ডিলিট করো	4	PASS
T10.v3	EN	can you cancel my first meeting on May 16	4	PASS
T10.v3	BN	16 মে তারিখে আমার প্রথম মিটিংটা বাতিল করে দিতে পারবে	4	PASS
T13.v1	EN	Cancel the next Team sync meeting	4	PASS
T13.v1	BN	পরের Team sync মিটিংটা বাতিল করো	4	PASS
T13.v2	EN	Can you cancel the next Team sync meeting	4	PASS
T13.v2	BN	পরের Team sync মিটিংটা বাতিল করে দিতে পারবে	4	PASS
T13.v3	EN	Delete the next Team sync meeting	4	PASS
T13.v3	BN	পরের Team sync মিটিংটা ডিলিট করো	4	PASS
T15.v1	EN	Cancel my next meeting with Dana	4	PASS
T15.v1	BN	Dana-এর সাথে আমার পরের মিটিংটা বাতিল করো	5	FAIL
T15.v2	EN	I need to cancel my next meeting with Dana. Can you do that for me please?	4	PASS
T15.v2	BN	Dana-এর সাথে আমার পরের মিটিংটা বাতিল করতে হবে। তুমি কি এটা করে দিতে পারবে, প্লিজ?	4	PASS
T15.v3	EN	Dana is off sick. Can you cancel my next meeting with them?	4	PASS
T15.v3	BN	Dana অসুস্থতার জন্য ছুটিতে আছে। ওর সাথে আমার পরের মিটিংটা বাতিল করে দিতে পারবে?	4	PASS
T16.v1	EN	If I haven't met with Carlos in the last 14 days, schedule a 30-minute meeting called 'catch-up' for my first free slot from tomorrow	4	PASS
T16.v1	BN	গত 14 দিনে যদি Carlos-এর সাথে আমার দেখা না হয়ে থাকে, তাহলে আগামীকাল থেকে আমার প্রথম ফাঁকা স্লটে 'catch-up' নামে একটা 30 মিনিটের মিটিং শিডিউল করো	4	PASS
T16.v2	EN	I think I might need to catch up with Carlos. Can you check if I've met with them in the last 14 days? If not, schedule a 30-minute meeting for my first free slot from tomorrow	4	FAIL*

T16.v2	BN	মনে হচ্ছে Carlos-এর সাথে একবার কথা বলা দরকার। একটু দেখো তো, গত 14 দিনে ওর সাথে আমার দেখা হয়েছে কি না? না হয়ে থাকলে আগামীকাল থেকে আমার প্রথম ফাঁকা স্লটে একটা 30 মিনিটের মিটিং শিডিউল করো	4	PASS
T16.v3	EN	have I met with Carlos in the last 14 days? If not, schedule a 30-minute meeting called 'catch-up' for my first free slot from tomorrow	4	FAIL*
T16.v3	BN	গত 14 দিনে কি Carlos-এর সাথে আমার দেখা হয়েছে? না হলে আগামীকাল থেকে আমার প্রথম ফাঁকা স্লটে 'catch-up' নামে 30 মিনিটের একটা মিটিং শিডিউল করো	4	PASS
T17.v1	EN	Cancel my meetings on Thursday after 14:00	5	PASS
T17.v1	BN	বৃহস্পতিবার 14:00টার পরে আমার মিটিংগুলো বাতিল করো	5	PASS
T17.v2	EN	Delete my meetings on Thursday after 14:00	5	PASS
T17.v2	BN	বৃহস্পতিবার 14:00টার পরে আমার যে মিটিংগুলো আছে, সেগুলো ডিলিট করো	5	PASS
T17.v3	EN	something came up. Can you cancel my meetings on Thursday after 14:00?	5	PASS
T17.v3	BN	হঠাৎ একটা কাজ পড়ে গেছে। বৃহস্পতিবার 14:00টার পরে আমার মিটিংগুলো বাতিল করে দিতে পারবে?	5	PASS
T18.v1	EN	Cancel all future meetings with Lee	5	PASS
T18.v1	BN	Lee-এর সাথে ভবিষ্যতের সব মিটিং বাতিল করো	5	PASS
T18.v2	EN	I need to cancel all future meetings with Lee. Can you do that for me please?	5	PASS
T18.v2	BN	Lee-এর সাথে সামনে যত মিটিং আছে, সবগুলো বাতিল করতে হবে। তুমি কি এটা করে দিতে পারবে, প্লিজ?	5	PASS
T18.v3	EN	Lee is leaving the company. Can you cancel all future meetings with them?	5	PASS
T18.v3	BN	Lee কোম্পানি ছেড়ে চলে যাচ্ছে। ওর সাথে ভবিষ্যতের সব মিটিং বাতিল করে দিতে পারবে?	5	PASS
T19.v1	EN	Cancel future Team sync meetings	7	PASS
T19.v1	BN	ভবিষ্যতের Team sync মিটিংগুলো বাতিল করো	5	PASS
T19.v2	EN	Delete all the future Team sync meetings	7	PASS
T19.v2	BN	ভবিষ্যতের সব Team sync মিটিং ডিলিট করো	5	PASS
T19.v3	EN	We've decided we don't need any any more Team sync meetings. Can you cancel all future ones?	5	PASS
T19.v3	BN	আমরা ঠিক করেছি, Team sync মিটিং আর লাগবে না। সামনে যতগুলো আছে, সব বাতিল করে দিতে পারবে?	5	PASS
T20.v1	EN	Create a 1.5 hour event called Roadmap review on May 22 at 14:00 with Dana	3	FAIL*
T20.v1	BN	22 মে 14:00টায় Dana-এর সাথে Roadmap review নামে একটা ইভেন্ট তৈরি করো, সময়কাল 1.5 ঘণ্টা	3	FAIL*
T20.v2	EN	I haven't met with Dana in a while. Can you schedule a 1.5 hour event called Roadmap review on May 22 at 14:00?	3	PASS
T20.v2	BN	অনেকদিন Dana-এর সাথে দেখা হয়নি। 22 মে 14:00টায় Roadmap review নামে 1.5 ঘণ্টা সময়ের একটা ইভেন্ট শিডিউল করে দিতে পারবে?	3	PASS
T20.v3	EN	I need to catch up with Dana. can you schedule a 1.5 hour event called Roadmap review on May 22 at 14:00?	3	PASS
T20.v3	BN	Dana-এর সাথে একটু বসা দরকার। 22 মে 14:00টায় Roadmap review নামে একটা ইভেন্ট শিডিউল করে দিতে পারবে? ইভেন্টটা 1.5 ঘণ্টা ধরে চলবে।	3	PASS
T54.v1	EN	If I haven't met with Priya in the past 7 days, send them an email titled 'Catch up soon?' saying 'We haven't caught up in a while - can you send some availability over next week?'	6	PASS

T54.v1	BN	গত 7 দিনে যদি Priya-এর সাথে আমার কোনো মিটিং না হয়ে থাকে, তাহলে তাকে “Catch up soon?” টাইটেল দিয়ে একটা ইমেইল পাঠাও, তাতে লেখো “We haven't caught up in a while - can you send some availability over next week?”	6	PASS
T54.v2	EN	I haven't met with Priya in a while. if it's been longer than 7 days can you send them an email titled 'Catch up soon?' saying 'We haven't caught up in a while - can you send some availability over next week?'	6	PASS
T54.v2	BN	Priya-এর সাথে অনেকদিন মিটিং হয়নি। যদি 7 দিনের বেশি হয়ে থাকে, তাহলে তাকে একটা ইমেইল পাঠাতে পারবে? টাইটেল হবে “Catch up soon?”, আর লেখা থাকবে “We haven't caught up in a while - can you send some availability over next week?”	9	PASS
T54.v3	EN	can't remember the last time I met with Priya. Can you check if it's been over 7 days? If so, send them an email titled 'Catch up soon?' saying 'We haven't caught up in a while - can you send some availability over next week?'	10	PASS
T54.v3	BN	শেষ কবে Priya-এর সাথে মিটিং করেছি, মনে পড়ছে না। একটু দেখবে, 7 দিনের বেশি হয়ে গেছে কিনা? হয়ে থাকলে তাকে “Catch up soon?” টাইটেল দিয়ে একটা ইমেইল পাঠাও, তাতে লেখো “We haven't caught up in a while - can you send some availability over next week?”	8	PASS
T55.v1	EN	If I don't have any meetings scheduled with Priya in the next 7 days, send them an email titled 'Catch up soon?' saying 'We have not caught up in a while - can you send some availability over next week?'	7	PASS
T55.v1	BN	আগামী 7 দিনে যদি Priya-এর সাথে আমার কোনো মিটিং শিডিউল করা না থাকে, তাহলে তাকে 'Catch up soon?' টাইটেল দিয়ে একটা ইমেইল পাঠাও, তাতে লেখো 'We have not caught up in a while - can you send some availability over next week?'	6	PASS
T55.v2	EN	Did I already schedule a meeting with Priya in the next 7 days? If not, send them an email titled 'Catch up soon?' saying 'We have not caught up in a while - can you send some availability over next week?'	6	PASS
T55.v2	BN	আগামী 7 দিনের মধ্যে Priya-এর সাথে আমি কি আগেই কোনো মিটিং শিডিউল করে রেখেছি? না করে থাকলে, তাকে 'Catch up soon?' টাইটেল দিয়ে একটা ইমেইল পাঠাও, তাতে লেখো 'We have not caught up in a while - can you send some availability over next week?'	5	PASS
T55.v3	EN	I need to check if I have any meetings scheduled with Priya in the next 7 days. If not, send them an email titled 'Catch up soon?' saying 'We have not caught up in a while - can you send some availability over next week?'	16	PASS
T55.v3	BN	আগামী 7 দিনে Priya-এর সাথে আমার কোনো মিটিং শিডিউল করা আছে কিনা, দেখতে হবে। না থাকলে, তাকে 'Catch up soon?' টাইটেল দিয়ে একটা ইমেইল পাঠাও, যেখানে লেখা থাকবে 'We have not caught up in a while - can you send some availability over next week?'	6	PASS